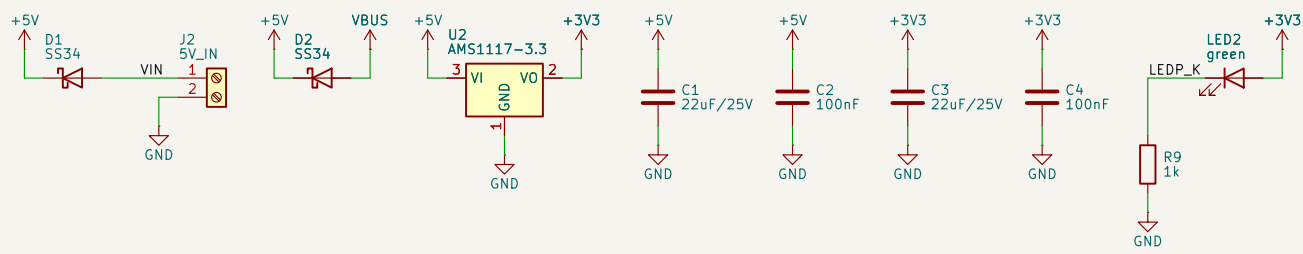
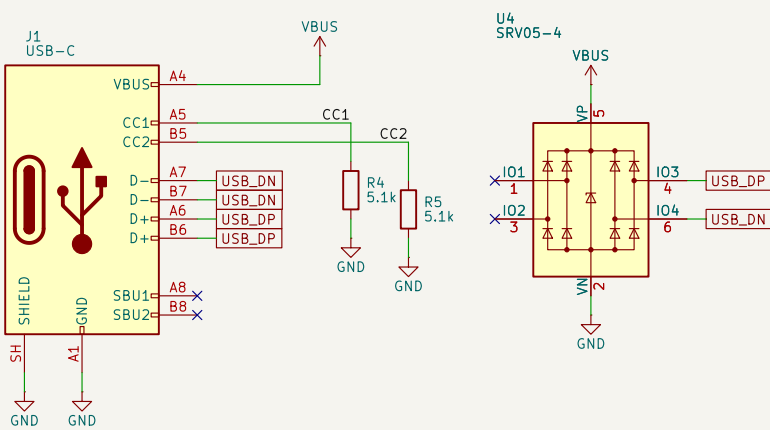


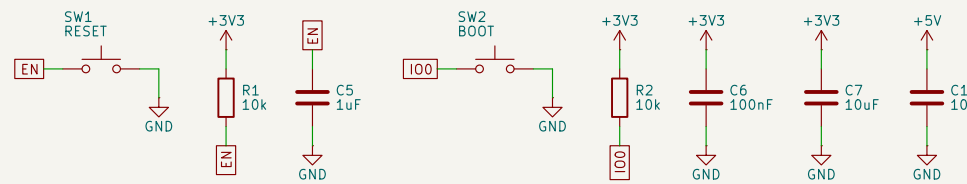
POWER IN
5V DC terminal or USB, ORed Schottky diodes



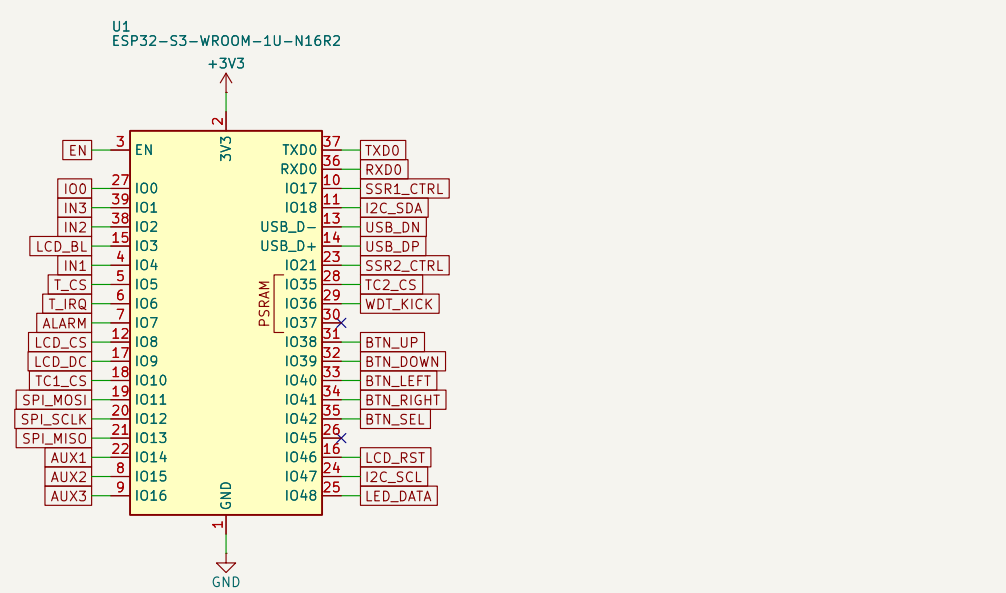
USB-C
native USB flashing + ESD



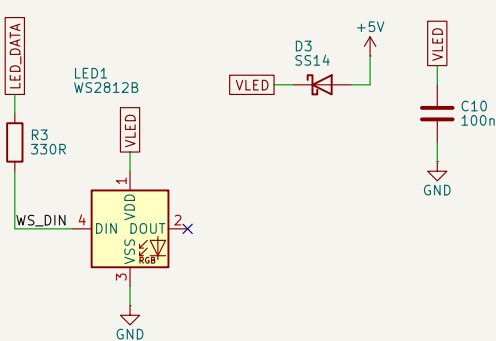
RESET / BOOT / DECOUPLING



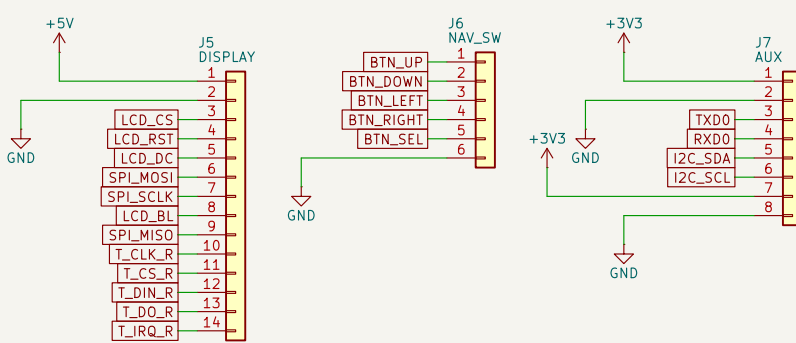
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



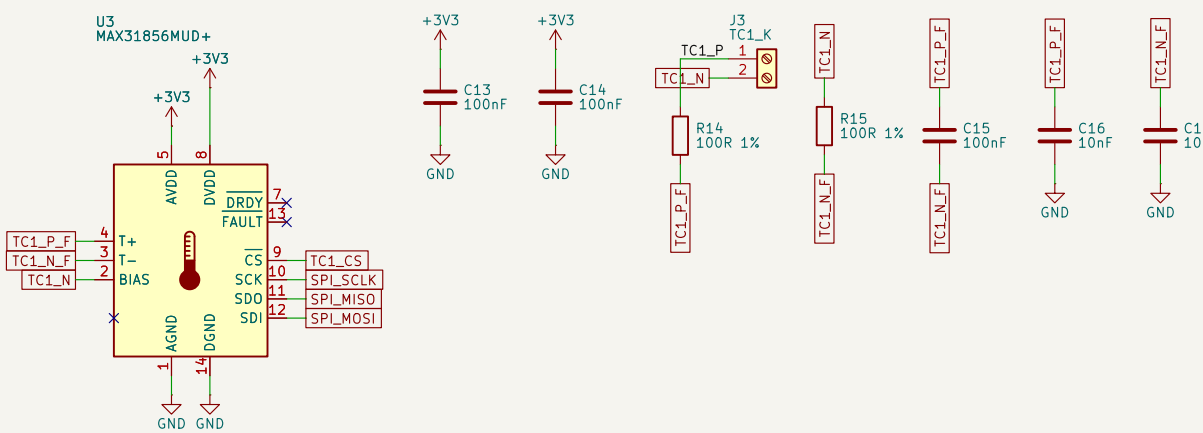
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



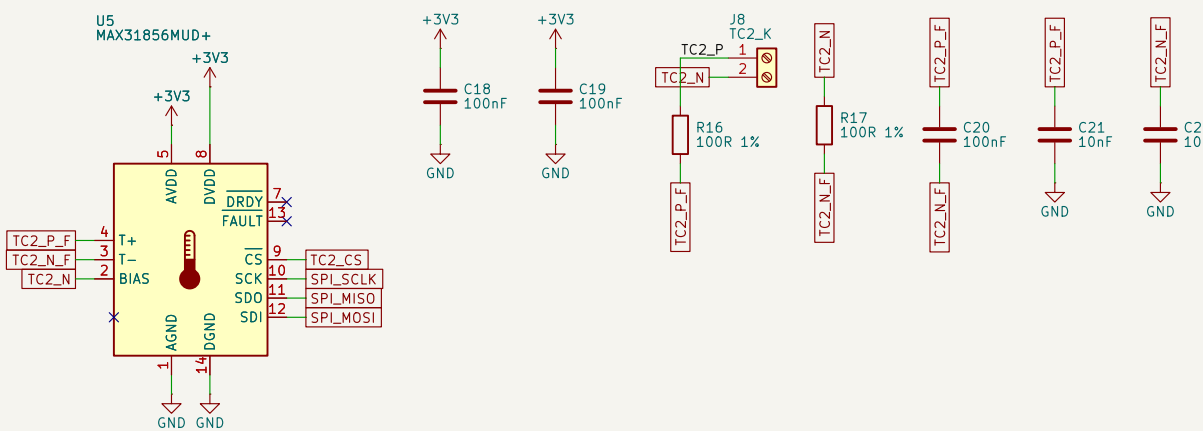
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



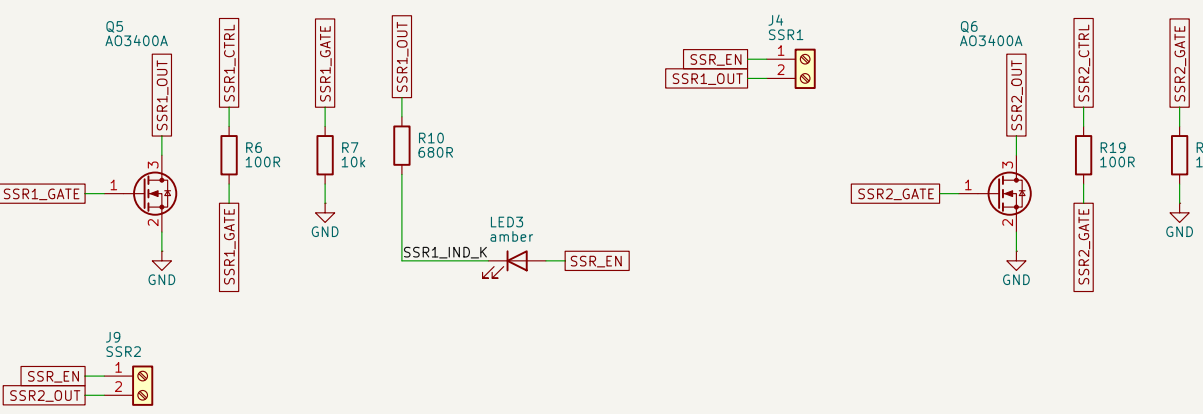
THERMOCOUPLE 1 (control) MAX31856
T- floats to J3, biased via BIAS



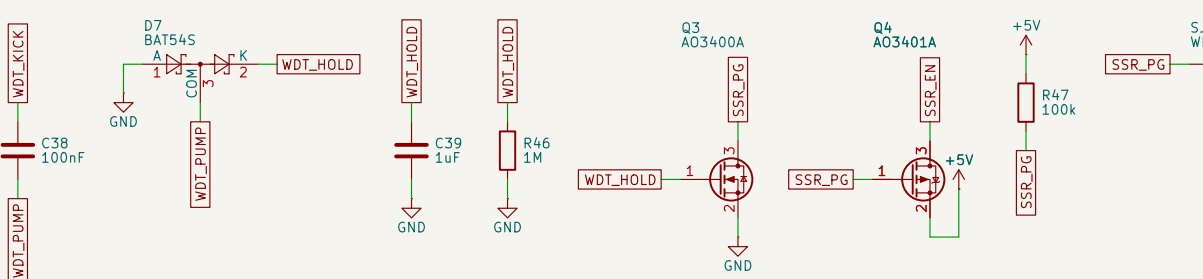
THERMOCOUPLE 2 (load) MAX31856
T- floats to J8, biased via BIAS



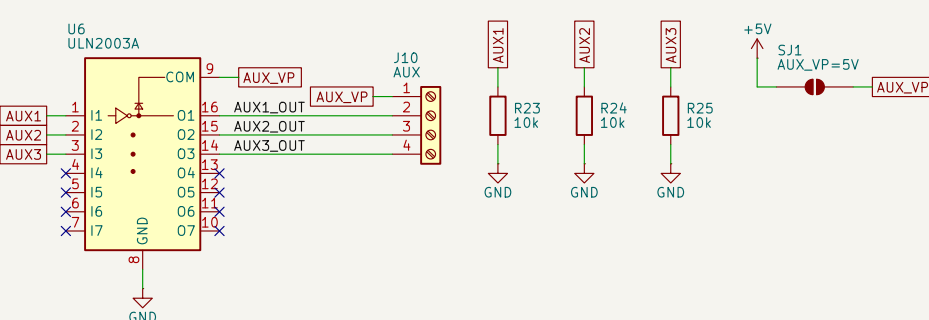
SSR DRIVE x2
low-side A03400A; indicator LED across each terminal



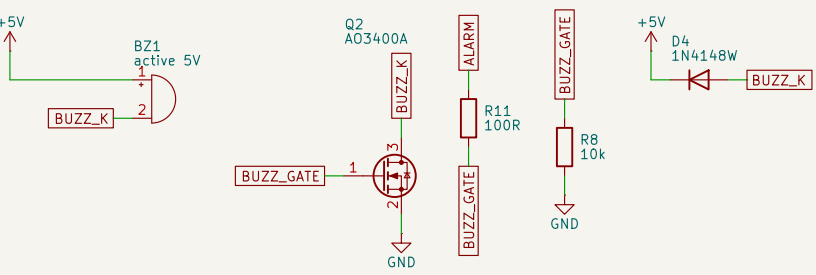
HARDWARE WATCHDOG
C38/D7/C39/R46 diode charge pump on WDT_KICK holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators. R47 is the fail-safe pull-up. SJ2 = bring-up defeat, REMOVE for service.



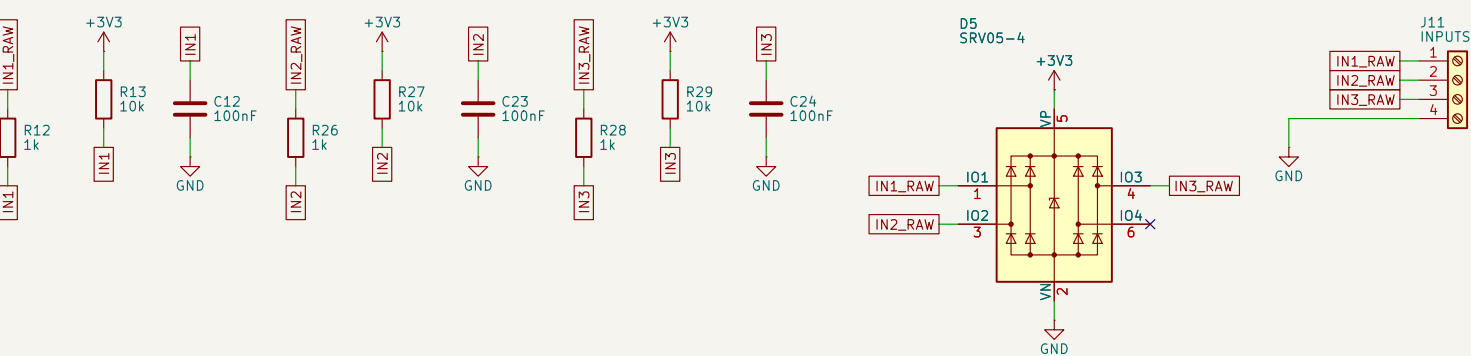
AUX OUTPUT BANK
ULN2003 vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



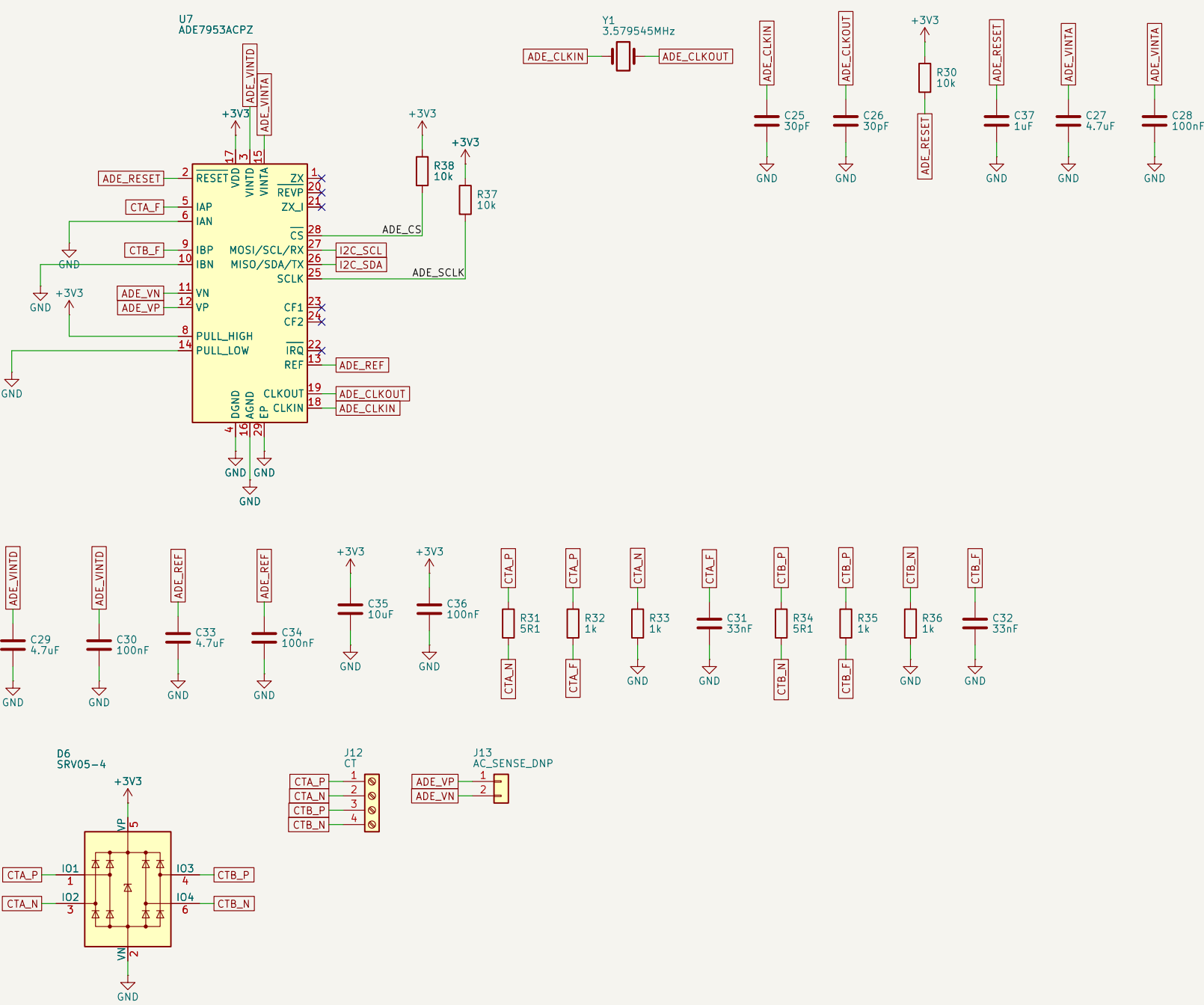
ALARM BUZZER



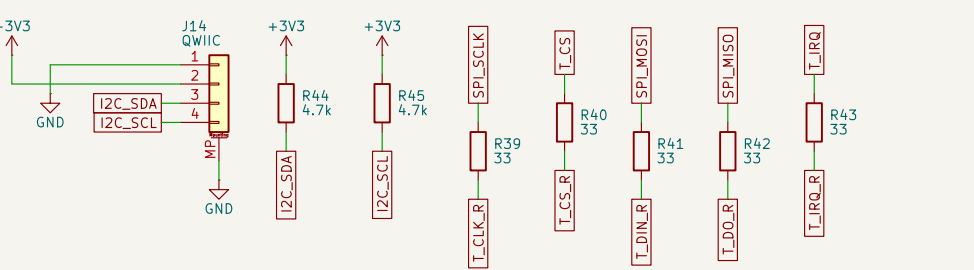
PROTECTED INPUTS x3 I_{lid} / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



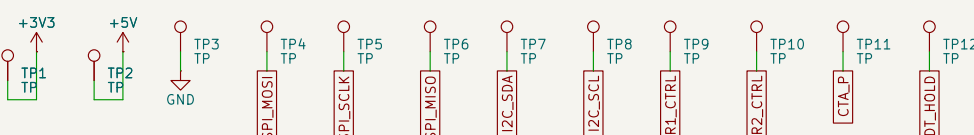
CT CURRENT SENSING ADE7953, I2C, current-only
no mains - VP/VN to DNP J13



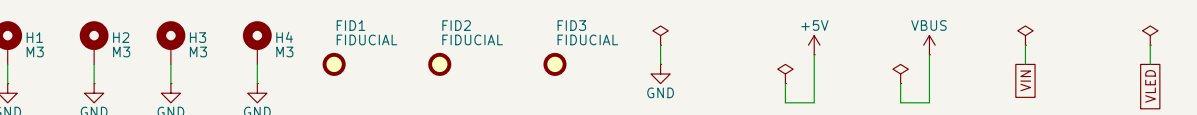
TOUCH DAMPING + I2C EXPANSION
R39-R43 damp the shared SPI2 bus for the display module's XPT2046 (not on this board). J14 Qwlic and J7 5-8 (0.1 in) share the I2C bus, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS
FID1-3 are bare-copper pick-and-place targets, on no net



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated), pin2 = the switched low side, Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated - rev B tried that and reversed it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal bias network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (I_{lid}/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST77965 + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RS 5=DC 6=SDI/MOSI 7=SSCK 8=BL 9=SDO/MISO 10=I2C_CLK 11=I2C_CS 12=I2C_D0 13=I2C_D1 14=I2C_D2 15=+5V NOT 3V3 - the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (ESP-R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N - one CT clamp per SSR zone.

J13: DNP SELV voltage sense header, NOT mains-rated, not fitted - do not wire to AC mains, I2/C25/C26 load caps are an ASSUMED value (unverified _L_); see design.py comment, ADE7953 I2C address 0x38 collides with PCT85746.